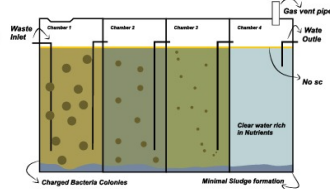


ZERO-WASTE ON-SITE BIO-SANITATION TECHNOLOGY

SMART BUDDY BIO-DIGESTER TANK (BDT)

Eco-Friendly Anaerobic Waste Treatment System with DRDO Inoculum Technology

SB-BDT Series (100L - 100kL)



Smart Buddy Bio-Digester Tank & 4-Chamber Digestion Principle

Technology & Process Overview

The Smart Buddy Bio-Digester is a zero-waste on-site sanitation treatment system using high-efficiency Anaerobic Microbial Inoculum (AMI bacteria). It completely eliminates the need for expensive sewerage networks, sewage treatment plants (STPs), or periodic septic tank suction cleaning. Waste is converted into odorless water and biogas naturally.

Key Environmental Highlights

- No sewerage network or STP plant dependency.
- Zero moving parts — completely maintenance-free operation.
- >99% Pathogen destruction & 99% Organic matter elimination.
- No de-sludging, vacuum suction, or manual scavenging needed.
- Effluent is 100% odorless, clear, and safe for gardening/irrigation.

DETAILED 4-CHAMBER WORKING MECHANISM & EFFLUENT PURIFICATION PROCESS



Chamber 1

Waste Inlet: High-density bacterial colonies initiate breakdown of organic solids.

Chamber 2

Liquefaction: Anaerobic fermentation converts solids into organic acids.

Chamber 3

Gas Venting: Methanogenesis releases methane & CO₂; minimal sludge remains.

Chamber 4

Water Outlet: Clear nutrient-rich effluent discharge safe for soil & plants.

Why Bio-Digester is Superior to Conventional Septic Tanks

1. Zero Maintenance & No Suction

Unlike septic tanks that need regular desludging, bio-digesters decompose 99% of solids automatically.

2. 100% Odorless & Fly-Proof

Closed anaerobic chamber process prevents foul odors, mosquitoes, flies, and cockroach infestation.

3. Pathogen & Disease Free

AMI bacteria destroy over 99.9% of harmful pathogens and fecal coliforms, preventing water contamination.

4. Reusable Liquid Effluent

Treated water contains valuable nitrogen and phosphorus nutrients, ideal for agricultural & garden irrigation.

Approved Green Sanitation Solution for Swachh Bharat Mission & Smart Cities

- Approved by DRDO & Ministry of Drinking Water and Sanitation
- Suitable for Extreme Cold & High Altitude
- Complies with CPCB & NGT Environmental Discharge Norms
- Available in FRP, HDPE & RCC Tank Modules

TECHNICAL SPECIFICATIONS: SMART BUDDY BIO-DIGESTER TANK (BDT)

TECHNICAL PARAMETER	ENGINEERING SPECIFICATION / FIELD STANDARDS
Digestion Technology	Anaerobic Bio-Digestion using Psychrotrophic Inoculum
Bacterial Formulation	Anaerobic Microbial Inoculum (AMI / DRDO Inoculum Bacteria)
Tank Construction	High-Density FRP (Fiber Reinforced Plastic) / Polyethylene / RCC Modular
Chamber Configuration	Multi-Stage 4-Chamber Internal Baffle Wall Segregation
Installation Options	Underground Installation (Zero Footprint) or Above Ground on RCC Pad
Pathogen Destruction	> 99.9% (Inactivates Typhoid, Cholera, Dysentery & Diarrhea pathogens)
Organic Matter Reduction	99% Total Suspended Solids (TSS) and Organic Sludge Breakdown
Treated Water Output	Clear, Odorless, Safe Non-Toxic Liquid Effluent (Rich in N-P-K plant nutrients)
Biogas Output	Methane (CH4) & Carbon Dioxide (CO2) vented safely or tapped for heating
Cleaning Requirement	Zero De-Sludging / No Manual Scavenging or Vacuum Suction required
Operating Temperature	Sub-Zero -20°C up to +55°C (Effective in Glaciers, High-Altitude & Deserts)
Toilet Cleaner Safety	Commercially available organic and mild toilet cleaners are fully compatible

Standard Capacity Sizing Guide (User Count vs Tank Volume)

Daily User Count	Tank Capacity (Litres)	Dimensions (L x W x H)	Ideal Application
5 - 10 Users	1,000 Litres	1500 x 900 x 900 mm	Individual Homes / Rural Houses
25 - 50 Users	2,500 Litres	2100 x 1200 x 1100 mm	Public Toilets / Small Offices
100 - 200 Users	5,000 Litres	3000 x 1400 x 1400 mm	Schools, Highway Petrol Pumps
500+ Users	10,000 - 50,000 L	Custom Modular Battery	Gram Panchayats, Hostels, Campuses

Recommended Installation Sectors & Application Areas

● Public & Community E2T Toilets	● Rural Gram Panchayats & Villages	● Highway Petrol Pumps & Dhabas	● Indian Defense & Border Posts
● Schools, Colleges & Hostels	● Hill Stations & Glaciers (-20°C)	● Tourist Spots & Heritage Parks	● Railway Stations & Bus Depots

HOW TO ENQUIRE & ORDER SMART BUDDY BIO-DIGESTER TANKS

Share your daily toilet user count, soil percolation conditions, and above/ underground site preferences. Our engineering team will calculate the exact retention volume, provide CAD piping layouts, and supply the Bio-Digester tank along with initial AMI bacterial inoculum charge.

DIRECT SALES & BIO-SANITATION ENQUIRY DESK:

- Direct Hotline: +91 88067 96868 / +91 99238 10197
- Email: sales@aaryainnovtech.com
- Web: www.aaryainnovtech.com
- Plant: S-27, Ambad MIDC, Nashik, MH - 422010